GTAP Practice Simulations

- What is the economic story?
- Which variable represents the shock/target?
- Is it EXO or ENDO under the current closure?
- Do we need a swap? Why?
- Which results tell us the economic mechanism—not merely whether the number is positive or negative?

S	Research Question	EXO.	ENDO.	Swap Requirement	What to Analyze
S 1	If GDP rises, must welfare rise? What about other economies				Real GDP (qgdp), exports (qxwreg), imports (qiwreg), terms of trade (tot), welfare (EV), and CO_2 emissions (gco2t).
S 2	Productivity in the U.S. Oil sector improves by 8%.			**But, What productivity improvement is required to make Oil production rise exactly 8%?**	Oil output (qo), oil price (ps), imports (qim), and bilateral exports (qxs).
S 3	What import-tax adjustment would be necessary to increase total EU Oil imports exactly 5%				Bilateral trade (qxs), EU imports (qim), U.S./EU oil output (qo), and emissions (gco2t).
S 4	EU imports of U.S. Oil must rise exactly 5%. What bilateral policy change is required?				The bilateral import-tax change tms; EU imports qim, EU/U.S. output qo, competing suppliers qxs, GDP qgdp, terms of trade tot, and welfare EV.
S 5	Carbon pricing becomes more stringent. gco2t("REGION") = RCTAXB("BLOC");				Regional CO_2 (gco2t), world CO_2 (gco2tw), energy output (qo), and GDP (qgdp)
S 6	Combined energy-security experiment: U.S. Oil output +8%; EU imports of U.S. Oil +5%; EU imports of U.S. Gas +11%; ROW aggregate Oil exports −3%.				GDP (qgdp), exports (qxwreg), imports (qiwreg), bilateral Oil/Gas (qxs), U.S. Oil output (qo), ROW Oil exports (qxw), prices (ps), terms of trade (tot), welfare (EV), and emissions (gco2t)/(gco2tw).

1 Key Terminologies for GTAP Simulation

Terminology	Definition	Interpretation in GTAP / RunGTAP
CGE / AGE Model	Computable General Equilibrium (CGE), also called Applied General Equilibrium (AGE), is a quantitative framework in which prices, production, consumption, trade, factor markets, income, and other markets are determined simultaneously.	A shock in one market can affect all other markets through prices, trade, factor allocation, income, and expenditure.
GTAP	The Global Trade Analysis Project provides a standardized global CGE framework, database, model, parameters, documentation, and software-related resources for applied policy analysis.	GTAP is not simply a database. It is an integrated global economic modeling framework.
GTAP Database	A benchmark representation of the world economy for a particular reference year, containing production, consumption, bilateral trade, factor payments, taxes, margins, and other economic flows.	It describes the initial equilibrium: $GTAPDatabase = \text{“}Whatis?''$
GTAP Model	The behavioral and accounting system that specifies how producers, households, governments, investors, and traders respond to changes in prices, income, technology, and policy.	It describes how the benchmark economy responds: $GTAPModel = \text{“}Howdoestheeconomybehave?''$
GTAP-E	An energy–environment extension of the GTAP framework that introduces detailed energy substitution, capital–energy relationships, CO_2 emissions, carbon taxation, and emissions trading.	Important for energy, climate, carbon-tax, oil, gas, and emissions policy simulations.
GEMPACK	General Equilibrium Modelling PACKage. A software system designed for constructing and solving large-scale CGE models.	GEMPACK performs the mathematical solution of the system underlying many GTAP applications.
RunGTAP	A graphical user interface for preparing, running, and examining GTAP simulations.	The researcher uses RunGTAP to choose the version, closure, shocks, solution method, and results.
Benchmark Equilibrium	The initial equilibrium represented by the GTAP database before a policy or economic shock is imposed.	All markets and accounting identities are mutually consistent at the benchmark: $F(X^0, Z^0; \theta) = 0.$
Counterfactual Equilibrium	The new equilibrium calculated after changing one or more exogenous variables.	The model solves: $F(X^1, Z^1; \theta) = 0.$ The difference between X^1 and X^0 represents the simulated effect.
Region	A country or aggregated group of countries represented separately in the model.	In the class aggregation, examples include `DVCUSA`, `DVCEU`, and `DVCROW`.
Commodity	A good or service produced, consumed, imported, or exported in the model.	Examples include Oil, Gas, Coal, Electricity, Agriculture, Transport, and other industries/services.

Terminology	Definition	Interpretation in GTAP / RunGTAP
Endowment	A primary factor of production owned by the regional household and supplied to productive activities.	Examples include: $Land,\ UnSkLab,\ SkLab,\ Capital,\ NatRes.$
Closure	The division of model variables into exogenous and endogenous components.	The closure determines the economic question being asked: $Closure = Whatisgivenversuswhatissolved?$
Swap	A closure operation that exchanges the exogenous/endogenous status of two conformable variable components.	Conceptually: $Target:\ ENDO \rightarrow EXO$ and $Instrument:\ EXO \rightarrow ENDO.$
CDE	Constant Difference of Elasticities demand system used for private household consumption in standard GTAP.	It permits non-homothetic household demand, so expenditure shares can change with income.
SLUG	A classification parameter identifying whether an endowment is mobile or sluggish across industries.	In the GTAP-E version used here: $SLUG(i) = 0 \Rightarrow mobile,$ while positive `SLUG` identifies sluggish factors.
General Equilibrium Effect	An indirect effect generated because a shock changes prices, production, trade, factor returns, income, and expenditure throughout the economic system.	A region does not need to receive a direct shock for its endogenous variables to change.
Sensitivity Analysis	Repeating a simulation using alternative parameter values or modeling assumptions to determine whether conclusions are robust.	For example, compare alternative values of: $ESUBD,\quad ESUBM,\quad ETRAE,\quad ELFKEN.$

2 The GTAP Simulation Logic in One Diagram

$$GTAPDATABASE$$
$$BenchmarkEconomy: \text{“}Whatis?''$$

$$\downarrow$$

$$GTAP/GTAP-EMODEL$$
$$Equations + BehavioralStructure$$

$$\downarrow$$

$$PARAMETERS$$
$$ESUBD,\ ESUBM,\ ETRAE,\ ELFENY,\ ELFKEN,\ldots$$

$$\downarrow$$

$$CLOSURE$$
$$WhichvariablesareEXOandwhichareENDO?$$

$$\downarrow$$

$$SHOCK$$
$$Whatchangesfromoutsidethemodel?$$

$$\downarrow$$

$$SWAP(ifrequired)$$
$$TargetbecomesEXO; InstrumentbecomesENDO$$

$$\downarrow$$

$$GEMPACK$$
$$Condense + Solve + Backsolve$$

$$\downarrow$$

$$COUNTERFACTUALEQUILIBRIUM$$

$$\downarrow$$

$$RESULTS$$
$$qgdp,\ qo,\ qxs,\ tot,\ EV,\ gco2t,\ldots$$

$$\downarrow$$

$$ECONOMICINTERPRETATION$$

S1:Swap qgdp("DVCUSA") = aoreg("DVCUSA"); Shock qgdp("DVCUSA") = 5; S2: Shock aoall("Oil","DVCUSA") = 8;
if Reverse S2B:Swap qo("Oil","DVCUSA")= aoall("Oil","DVCUSA"); Shock qo("Oil","DVCUSA") = 8; S3:Swap qim("Oil","DVCEU")
= tm("Oil","DVCEU"); Shock qim("Oil","DVCEU") = 5; S4:Swap qxs("Oil","DVCUSA","DVCEU")= ams("Oil","DVCUSA","DVCEU"
Shock qxs("Oil","DVCUSA","DVCEU") = 5; S5: Swap gco2t("REGION")= RCTAXB("BLOC"); Shock gco2t("REGION") =
-10; S6:Swap qo("Oil","DVCUSA") = aoall("Oil","DVCUSA"); Swap qxs("Oil","DVCUSA","DVCEU") = tms("Oil","DVCUSA","DVCE
Swap qxs("Gas","DVCUSA","DVCEU") = tms("Gas","DVCUSA","DVCEU"); Shock qo("Oil","DVCUSA") = 8; Shock qxs("Oil","DVC
= 5; Shock qxs("Gas","DVCUSA","DVCEU") = 11; S6B: Shock aoall("Oil","DVCUSA") = 8; Shock ams("Oil","DVCUSA","DVCEU")
= 5; Shock ams("Gas","DVCUSA","DVCEU") = 11; Shock tx("Oil","DVCROW") = [chosen policy shock];